

Assignment3 BS

Group 2

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1 Part I

1.1 Risk aversion parameter calibration

Using the initial full estimation window, we calibrated the risk aversion parameter (γ) for the mean-variance portfolio to target a gross exposure of approximately 3. By evaluating the portfolio's sensitivity to various levels of risk aversion, we determined that a value of $\gamma = 158$ satisfies the target leverage.

Moreover, we did closed-form checks on the 2014/12/31 rebalance window (46 assets).

	in_sample_mean_return	in_sample_volatility	gross_exposure
Minimum Variance	0.000468	0.006151	2.777753
Mean-Variance	0.000851	0.006345	3.000311

The linear shrinkage estimator is defined as:

$$\hat{\Sigma}_{\text{Shrink}} = \delta F + (1 - \delta)S \quad (1)$$

where:

- S is the sample covariance matrix (unstructured, noisy).
- F is the shrinkage target (structured, stable).
- $\delta \in [0, 1]$ is the shrinkage intensity.

1.2 Rolling Volatility

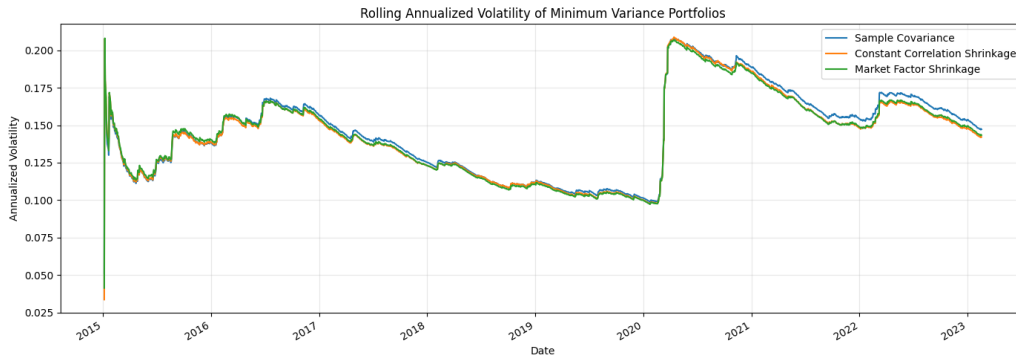


Figure 1: Rolling Volatility - Minimum Variance Ptf

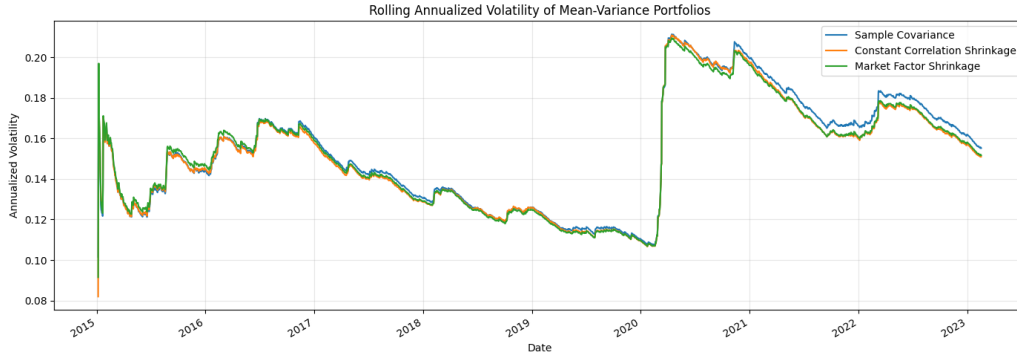


Figure 2: Rolling Volatility - Mean Variance Ptf

The rolling annualized volatility results demonstrate that both Constant Correlation and Market Factor shrinkage slightly outperform the Sample Covariance by providing lower and more stable out-of-sample risk profiles. However the merits of the shrinkage are not fully revealed in this plot due to the reasons explained in the last section.

1.3 Cumulative Returns

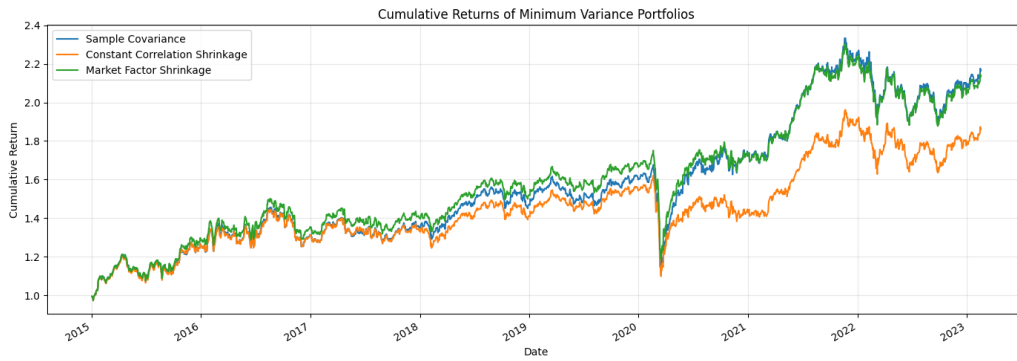


Figure 3: Cumulative Returns - Minimum Variance Ptf

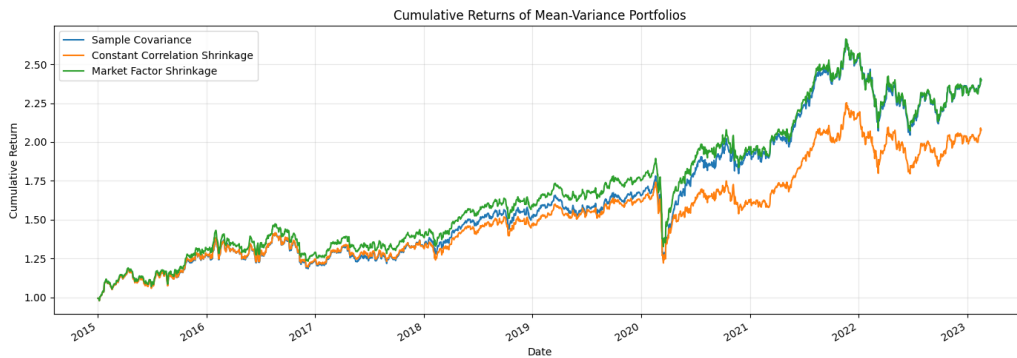


Figure 4: Cumulative Returns - Mean Variance Ptf

We can notice that in both strategy types, all estimators follow a similar trajectory, capturing the general market uptrend from 2015 to 2023. The market factor shrinkage is the top performer while the Constant Correlation target underperforms in both cases. While Constant Correlation is a robust theoretical target, its underperformance here suggests it may be too rigid for this specific investment universe.

1.4 Turnover

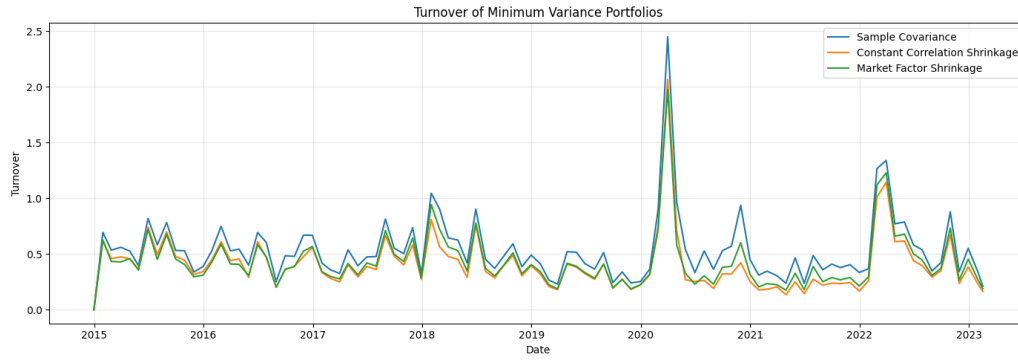


Figure 5: Turnover - Minimum Variance Ptf

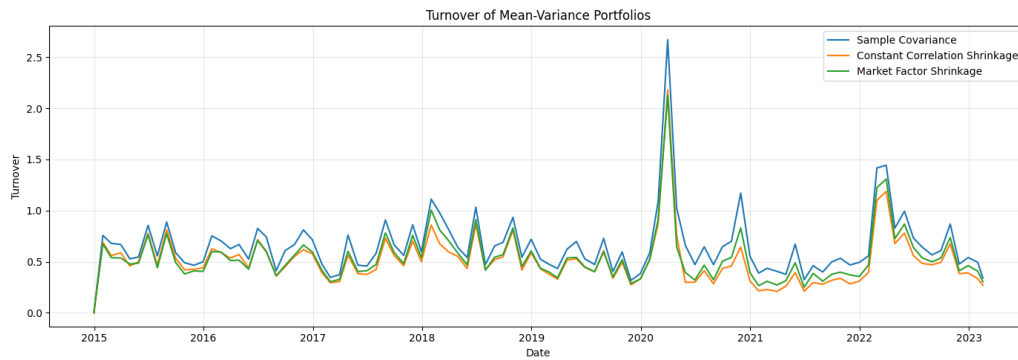


Figure 6: Turnover - Mean Variance Ptf

Methodology	Avg Turnover - Minimum Variance	Avg Turnover - Mean Variance
Sample Covariance	0.53	0.65
Constant Corr Shrinkage	0.41	0.51
Market Factor Shrinkage	0.43	0.54

Table 1: Average Turnover across different Covariance Estimation Models

Table 2 confirms that Sample Covariance produces the highest turnover (0.53), meaning it requires more frequent trading. This happens because the sample matrix is unstable and reacts to market noise. Shrinkage methods (0.41 and 0.43) are more stable and significantly reduce the need for costly rebalancing.

1.5 Gross Exposure

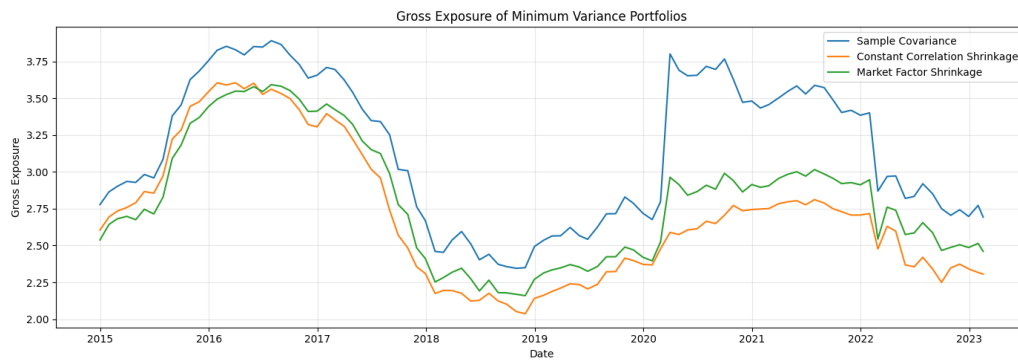


Figure 7: Gross Exposure - Minimum Variance Ptf

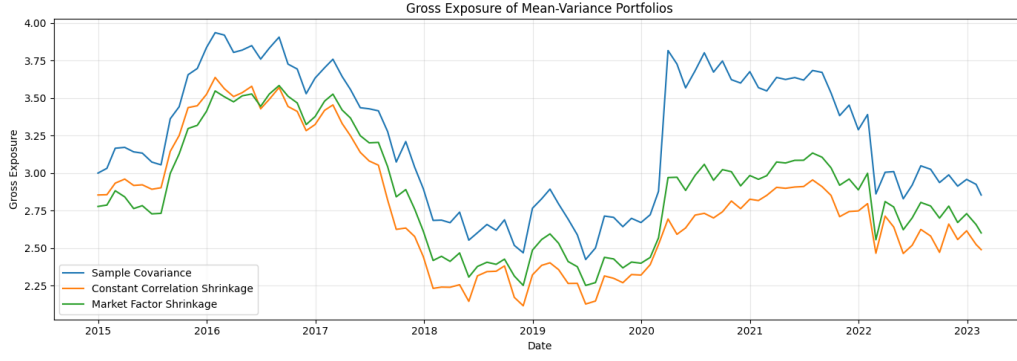


Figure 8: Gross Exposure - Mean Variance Ptf

Methodology	Gross Exposure - Minimum Variance	Gross Exposure - Mean Variance
Sample Covariance	3.14	3.22
Constant Corr Shrinkage	2.70	2.77
Market Factor Shrinkage	2.81	2.88

Table 2: Average Gross Exposure across different Covariance Estimation Models

The Sample Covariance model yields the highest gross exposure (3.14) because it reacts too much to noise, creating extreme and unstable long-short positions. Shrinkage methods effectively reduce this leverage to more realistic levels, with Constant Correlation being the most conservative (2.70).

1.6 Intensity

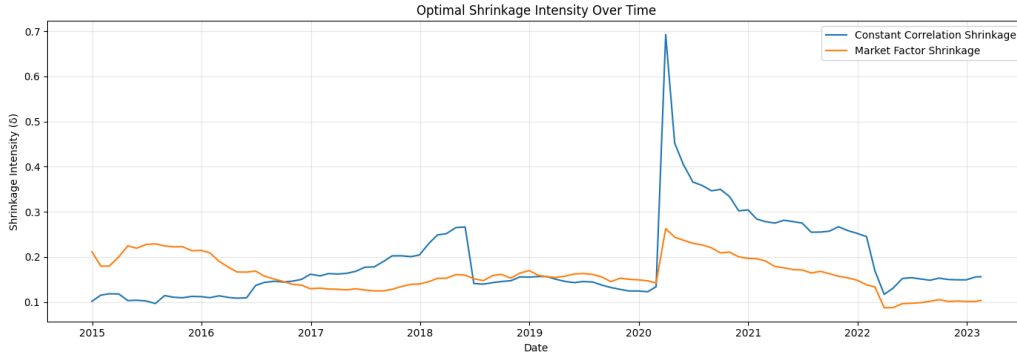


Figure 9: Optimal Shrinkage Intensity Over Time

The optimal shrinkage intensity δ balances the noisy sample covariance S with the structured target F . In particular, it's approximately defined by the ratio $\frac{1}{T} \frac{\pi - \rho}{\gamma}$, where π represents the estimation error in the sample covariance S , ρ represents the asymptotic covariance between the target F and the sample S , and γ represents the misspecification (bias) of the target F .

The empirical results in the chart clearly reflect these dynamics. During the 2020 market crisis, the sharp spike in both lines is driven by a surge in π , the estimation error. In high-volatility regimes, the sample entries become extremely noisy, making the unregularized S unreliable and forcing the intensity δ higher to maintain stability. The Constant Correlation model reaches a peak intensity of nearly 0.7, indicating that 70% of the portfolio's risk structure was derived from the average correlation target. Because this target is highly rigid and structurally simple, it requires a massive pull to override the noise (π) generated during the crash.

In contrast, the Market Factor intensity peaked significantly lower, at approximately 0.25. From a mathematical perspective, since the target F is a better "fit" for the observed market reality than a simple average, the estimator does not need to shrink as aggressively to find the optimal balance. This allows the Market Factor model to remain more trusting of the sample data, filtering out idiosyncratic noise while retaining the meaningful market-driven signals that are essential for accurate risk management.

1.7 Condition Number

The Condition Number measures the sensitivity of the covariance matrix to small perturbations in input data; a lower value indicates a more stable and "well-conditioned" matrix for optimization. The Sample Covariance matrix consistently exhibits the highest condition number, which explains the instability and high turnover observed in previous charts. Shrinkage methods significantly improve numerical stability by reducing this ratio, with Constant Correlation providing the most robust results. This reduction in the condition number is directly responsible for the lower turnover and more realistic gross exposure levels achieved by the regularized models.

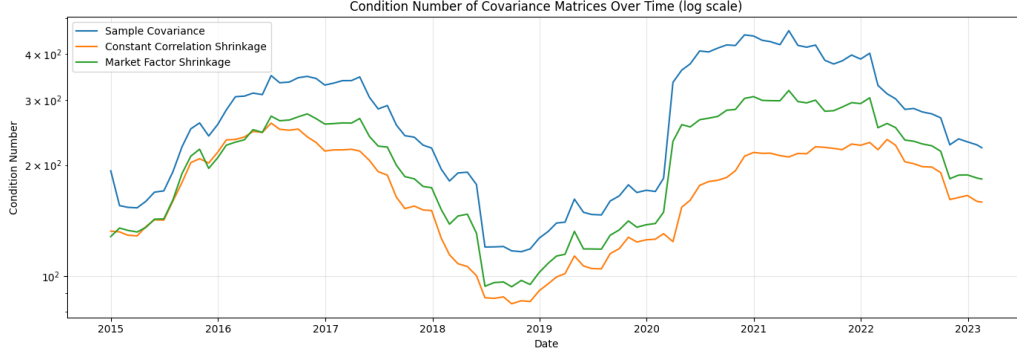


Figure 10: Condition Number of Covariance Matrices Over Time

2 Part II

2.1 Factor Turnover and Stability: Cosine Similarity Analysis

To quantify the stability of the pervasive risk sources identified through Principal Component Analysis (PCA), we analyze the "factor turnover". This is achieved by measuring the Cosine Similarity between the principal components obtained in consecutive rebalancing periods.

2.1.1 Theoretical Framework

The Cosine Similarity between two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as the dot product of the vectors normalized by the product of their magnitudes:

$$\text{Cos}_{\text{similarity}}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad (2)$$

2.1.2 Numerical Implementation

The implementation utilizes vectorized operations to compute the proximity matrix $\mathbf{P} = \mathbf{A}^\top \mathbf{B}$. We applied the Cosine distance formula by normalizing the matrices' columns (which represent the principal components weights):

```
a_normed = a / np.linalg.norm(a, axis=0)
b_normed = b / np.linalg.norm(b, axis=0)
pc_proximity = a_normed.T @ b_normed
```

The resulting matrix $\mathbf{P}$ has dimensions $(K \times K)$, where the element $p_{i,j}$ represents the similarity between the i -th current principal component and the j -th previous principal component.

Two primary metrics are derived from this matrix:

1. **Direct Similarity (Diagonal):** $\text{diag}(\mathbf{P})$ measures the persistence of each specific factor over time. A value close to 1 suggests that the principal component has remained stable in its composition.
2. **Maximum Absolute Similarity:** Computed as the maximum value across each row of $|\mathbf{P}|$. When compared to the diagonal similarity, this metric identifies if a principal component has maintained its economic identity while shifting its ordinal rank. A high maximum similarity coupled with a low diagonal value suggests that the risk driver has not been replaced, but its relative importance in explaining total variance has changed due to more (or less) specific market movements.

2.1.3 Percentage of Variance and Market Conditions

The analysis of the eigenvalue spectrum across the rolling windows reveals that the proportion of total variance explained by the principal components is a dynamic measure reflecting the prevailing market regime. As illustrated in the provided plot, there is a distinct inverse relationship between the variance explained by the first principal component only (PC1) and the number of components required to reach a 75% cumulative variance threshold.

Under "calm" market conditions, such as those observed between late 2018 and early 2020, the variance is distributed across multiple factors. During this period, PC1 explained a lower portion of the total risk, while the proportion of PCs needed to explain the 75% threshold reached its local maximum, suggesting that stock returns were driven by idiosyncratic sources, providing a higher-dimensional risk space.

Conversely, during periods of systemic market stress, we observe a significant concentration of risk. A prominent example in the plot is the sharp spike in PC1 variance explanation at the onset of 2020, where PC1 suddenly surged from below 40% to over 55%. Simultaneously, the dimensionality of the risk space collapsed, with the proportion of PCs needed to explain the 75% threshold dropping precipitously to nearly 15%.

The empirical results confirm that under high volatility the market becomes increasingly monodimensional, driven by a primary pervasive source of risk.

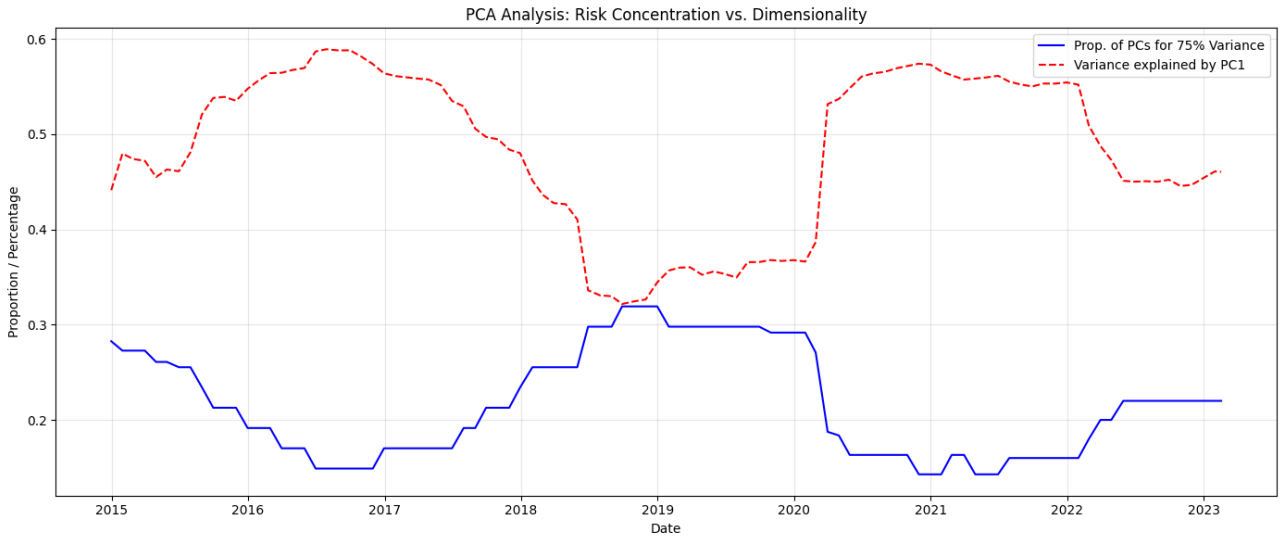


Figure 11: PCA Analysis: Risk Concentration (PC1) vs. Dimensionality (75% Threshold)

2.2 Empirical Analysis of Factor Turnover

To further investigate the pervasive sources of risk, we examine the "factor turnover" through the evolution of Cosine Similarity for the first four Principal Components (PCs). This analysis addresses the stability of the statistical factors over time and their sensitivity to the estimation window.

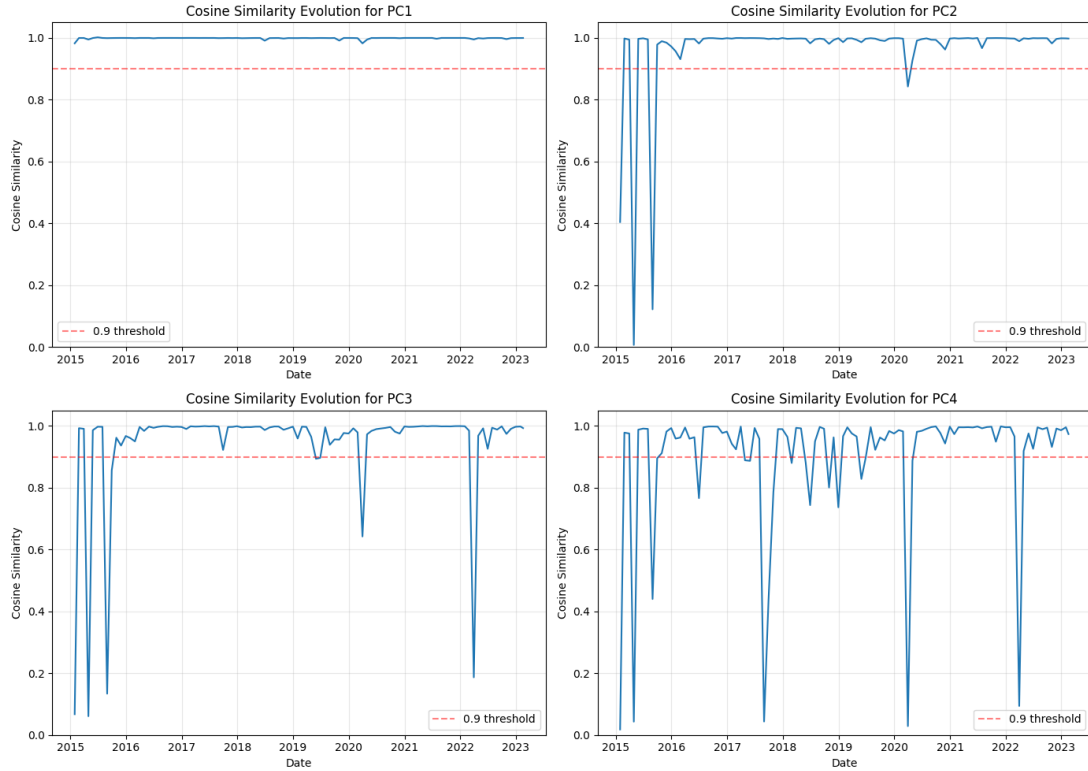


Figure 12: Cosine Similarity evolution for PC1 through PC4. The red dashed line represents the 0.9 stability threshold.

2.2.1 Results and Interpretation

The results displayed in Figure 12 highlight a clear hierarchy in factor stability:

- **PC1 Stability:** The first principal component (Market Factor) exhibits near-perfect stability, with similarity values consistently close to 1.0. This indicates that the primary driver of market risk remains structurally identical across rebalancing periods.
- **Higher-Order Components (PC2-PC4):** As the analysis moves beyond the first factor, the similarity becomes increasingly volatile. This reflects the fact that higher-order components capture more granular, idiosyncratic information rather than broad market trends.
- **Factor Reordering:** The sharp spikes toward zero in PC3 and PC4, followed by immediate recovery, often indicate "eigenvalue flipping," where two factors with similar explained variance swap their ordinal positions.

2.2.2 Discussion on Estimation Windows

The factor turnover is intrinsically linked to the rolling window size (24 months). A shorter window would increase the sensitivity to recent market noise, leading to higher turnover and lower cosine similarity. Conversely, our 2-year lookback period aims to capture structural risk relationships while remaining responsive to significant changes in market conditions.

2.2.3 PCA via Correlation Matrix

In this section, we performed the Principal Component Analysis (PCA) on the **correlation matrix** by standardizing the returns (i.e., dividing each return series by its standard deviation). The financial rationale behind this approach is to eliminate the idiosyncratic variance of individual assets focusing on the co-movement structure.

Factor Turnover and Stability As shown in Figure 13, the factor turnover is noticeably lower compared to the covariance-based approach.

This increased stability occurs because the correlation matrix filters out temporary "noise" or idiosyncratic spikes in individual stock volatility that could otherwise force an artificial rotation of the principal components. By focusing on pure correlations, the risk directions remain more consistent over the 24-month rolling windows.

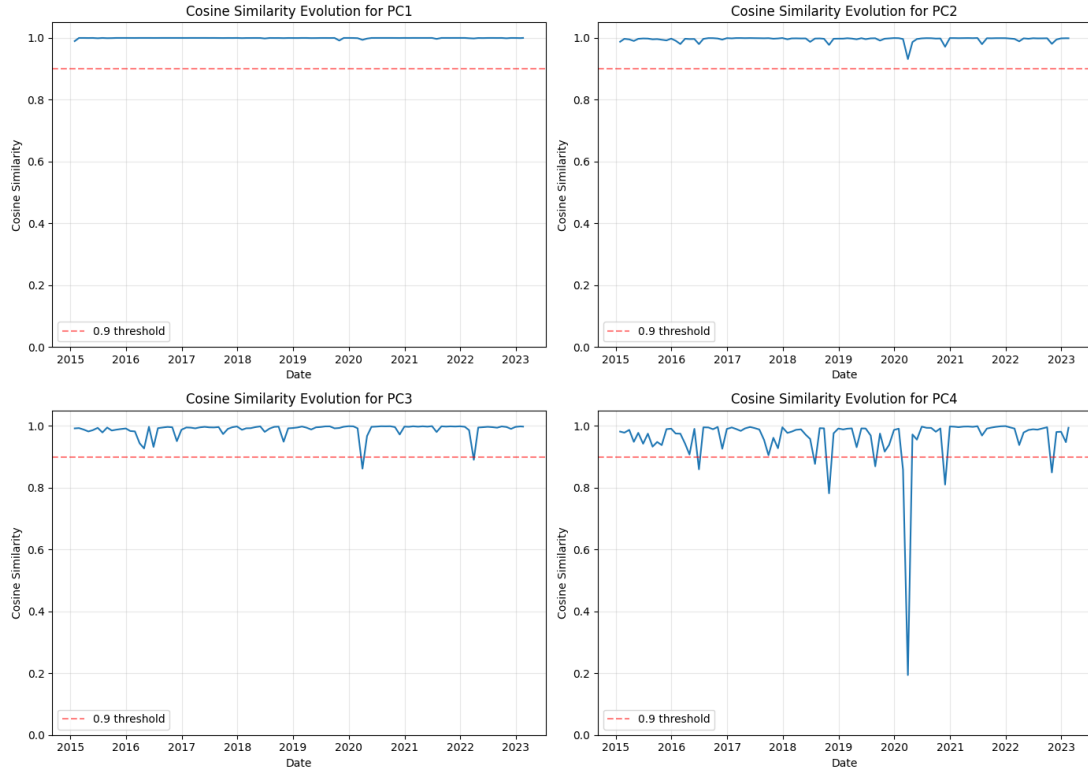


Figure 13: Cosine Similarity Evolution (Correlation-based).

Conversely, the plot for explained variance versus PC1 remained unchanged; thus, it has been omitted for brevity.

Comparative Statistical Analysis The following table compares the loadings for the first three principal components under both methods:

Metric	Covariance Based			Correlation Based		
	PC1	PC2	PC3	PC1	PC2	PC3
Mean	0.1395	-0.0058	0.0305	0.1429	0.0095	-0.0091
Min	0.0532	-0.3231	-0.6398	0.0613	-0.2407	-0.3072
Max	0.2575	0.3768	0.2074	0.1904	0.2911	0.2542
Std	0.0430	0.1473	0.1442	0.0295	0.1471	0.1472

Table 3: Descriptive Statistics of Loadings: Covariance vs. Correlation PCA

The data shows that for **PC1**, which represents the Market Factor, the correlation-based approach yields a significantly lower standard deviation (0.0295 vs. 0.0430) and extreme values are closer. This compression indicates that the loadings are more homogeneous, better reflecting the collective correlation with the market index rather than the idiosyncratic risk of high-volatility constituents.

2.3 Interpretation of the first three principal components

To interpret the first three principal components on January 31st, 2019, we analyze the average loadings across sectors.

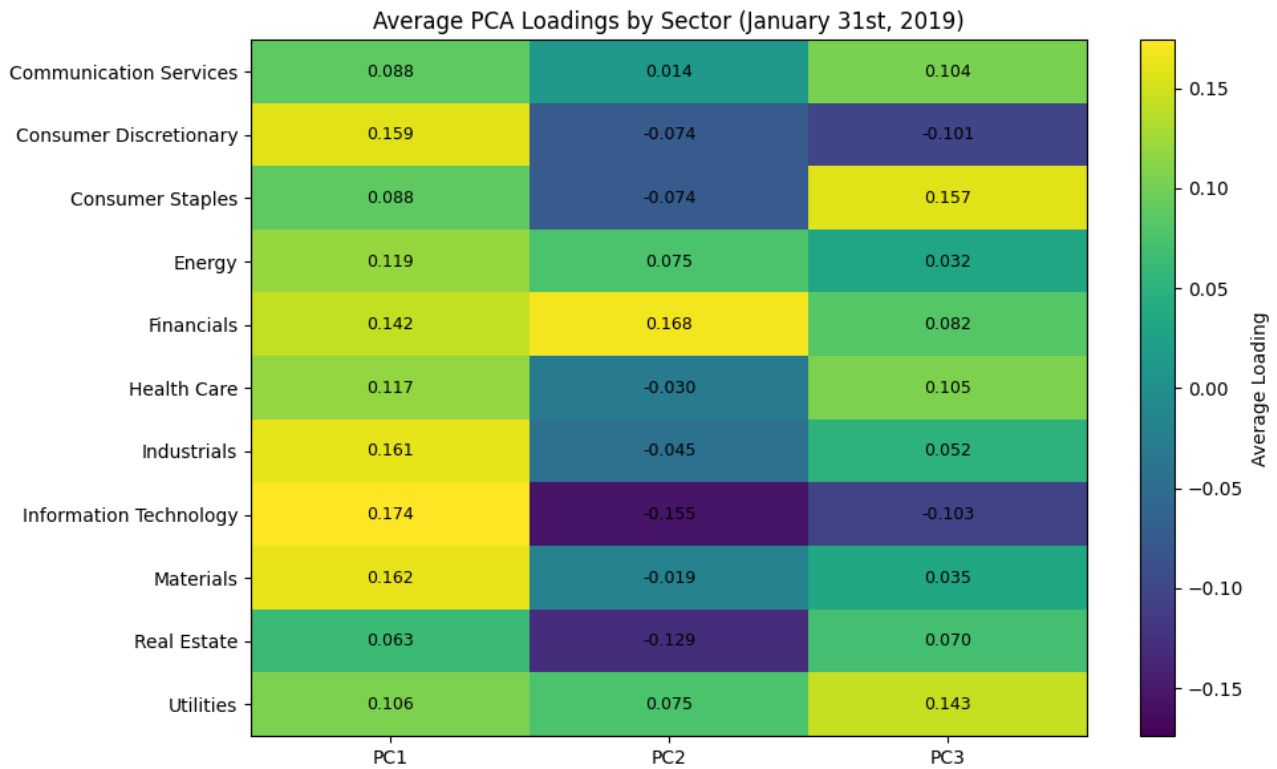
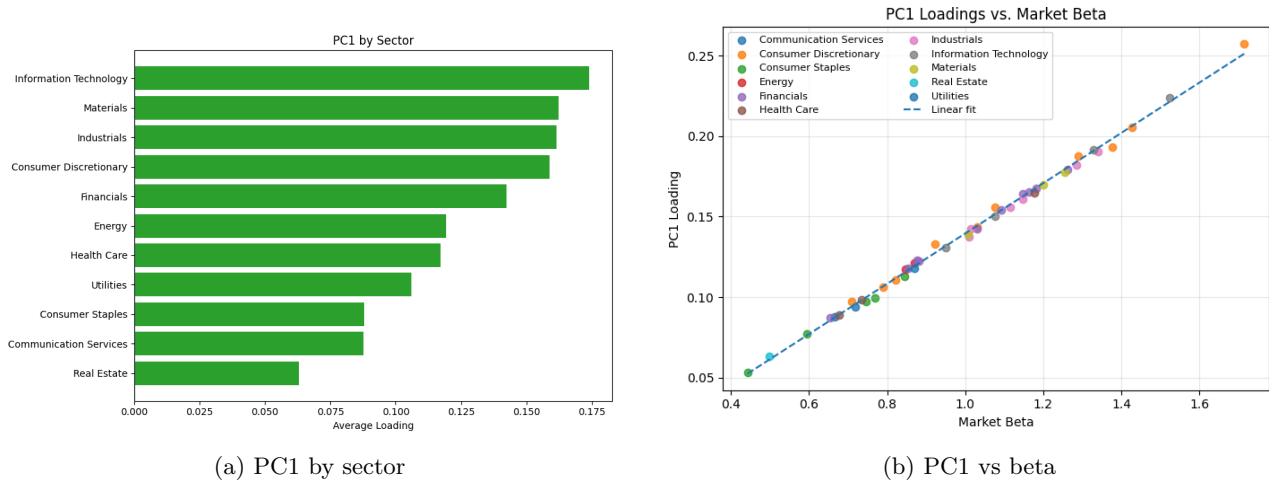
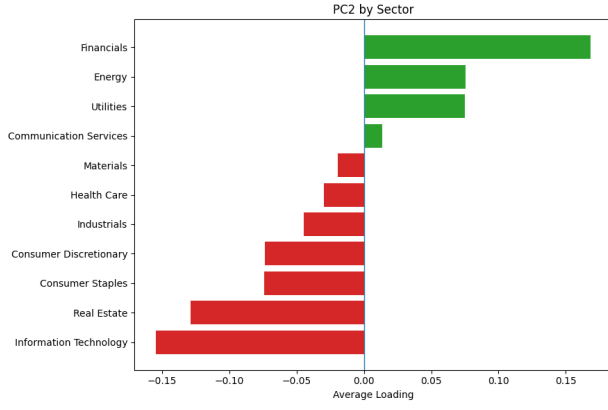


Figure 14: PCA loadings by sector

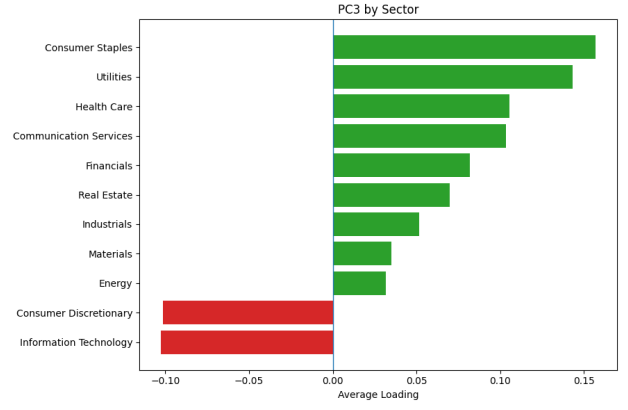
The heatmap highlights a clear structure in the cross-sectional distribution of loadings. PC1 is uniformly positive across all sectors, indicating a common source of variation. In contrast, PC2 and PC3 exhibit pronounced sign differences, suggesting the presence of sector-specific and style-related factors.



The PC1 loadings are positive and relatively homogeneous across sectors, confirming its interpretation as a market-wide factor. This is further supported by the strong linear relationship between PC1 loadings and market betas, indicating that this component captures the overall exposure to market risk.



(a) PC2 by sector



(b) PC3 by sector

PC2 displays a clear separation between sectors with positive and negative loadings. This pattern reflects a sector rotation effect, where different groups of industries respond differently to underlying economic conditions. PC3 further differentiates the cross-section by contrasting defensive and growth-oriented sectors.

3 Part III

3.a: connection between shrinkage methods and PCA-based denoising of the covariance matrix

In this section, we analyze the relationship between shrinkage estimators and PCA-based denoising techniques by studying the behavior of the eigenvalue spectrum of the covariance matrix. The analysis is conducted on the estimation window ending on January 31st, 2019.

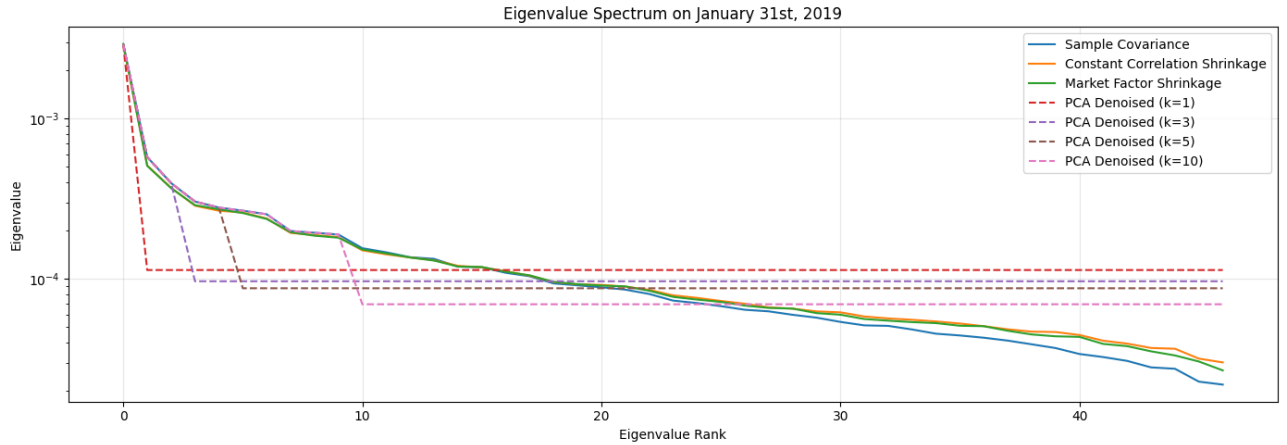


Figure 17: Eigenvalue spectrum of the covariance matrix on January 31st, 2019

As the plot shows, the eigenvalue spectrum of the sample covariance matrix exhibits a typical pattern characterized by a few large eigenvalues followed by a long tail of smaller ones.

PCA-based denoising modifies this structure by explicitly separating signal and noise. In particular, the first k eigenvalues are preserved, while the remaining ones are replaced with their average level. This results in a flattened tail of the spectrum, reflecting the assumption that all residual components beyond the first k principal components represent homogeneous noise. As k increases, a larger portion of the original spectrum is retained, progressively reducing the degree of denoising.

Shrinkage estimators, on the other hand, act differently. Instead of imposing a sharp separation between signal and noise, they perform a smooth regularization of the entire covariance matrix. This is reflected in the eigenvalue spectrum by a compression of extreme values: large eigenvalues are slightly reduced, while small eigenvalues are increased. As a result, the dispersion of the spectrum is reduced without enforcing a low-rank structure.

3.b: limitations due to the limited cross-sectional dimension

The dataset used in this analysis does not fully highlight the benefits of shrinkage techniques due to its limited cross-sectional dimension.

Shrinkage methods are particularly effective in high-dimensional settings, where the number of assets is large relative to the number of observations. In such cases, the sample covariance matrix is poorly conditioned and highly sensitive to estimation noise, giving shrinkage a much more significant role in improving stability, reducing estimation error, and enhancing out-of-sample performance.

In our case, the number of assets is relatively small, while the estimation window is sufficiently long. As a result, the sample covariance matrix is already reasonably well-behaved, and the impact of shrinkage is less pronounced, both in terms of eigenvalue spectrum and portfolio performance.

Code Errors

Utilities:

`constant_correlation_shrinkage:`

- `std_devs = np.sqrt(variances)`
- `diff = outer_prod - S`
- `off_diag_contrib = (avg_corr * (sqrt_ratio_matrix.T * theta_ii_ij + sqrt_ratio_matrix * theta_jj_ij) / 2)`
- `pi_hat /= T` and `rho_hat /= T` have been put outside the for loop.
- `denominator = gamma_hat * T`
- `"shrunk_cov": target * intensity + S * (1 - intensity)`

`market_factor_shrinkage:`

- `market_variance = market_aligned.var()`
- `diff = outer_prod - S`
- `pi_hat += np.sum(diff**2)`
- `pi_hat /= T`
- `"shrunk_cov": (1 - intensity) * cov_matrix + intensity * target`

`_align_portfolios_and_returns:`

- `aligned_portfolios = portfolios.loc[:, overlapping_assets].reindex(returns.index).ffill().shift().dropna(axis=0, how="all")`

Notebook:

- `time_horizon = 2 * 252`